📡 Ethernet Lan Switching and Address Resolution Protocol Part 2 (10.24.2026)

Source: Jeremy's IT Lab – CCNA 200-301

Layer: Layer 2 / Data Link

Main topics: Ethernet frames, ARP, ARP table, Ping/ICMP, MAC address table, switch forwarding behavior

1. Video Roadmap

Timestamp	Topic
0:57	Ethernet frames
7:41	ARP Request
10:15	ARP Reply
12:04	ARP table
13:05	GNS3 demo
15:31	Ping demo
18:28	Wireshark capture
20:25-22:10	MAC address table + clearing entries
26:00	Summary
27:06-31:58	Quiz questions

2. Ethernet Frame

> **Ethernet frame = Layer 2 PDU.** It encapsulates the Layer 3 packet inside the Ethernet payload.

Basic Ethernet II frame fields

Field	Purpose
Preamble / SFD	Synchronization / start of frame
Destination MAC	Receiving NIC/interface MAC
Source MAC	Sending NIC/interface MAC
Type/Length	Usually identifies Layer 3 protocol, e.g. IPv4 `0x0800`, ARP `0x0806`
Payload	Encapsulated L3 packet, e.g. IP packet
FCS	Frame Check Sequence – error detection trailer

Important frame-size facts

- **Minimum Ethernet payload size = 46 bytes**
- **Minimum Ethernet frame size = 64 bytes** from Destination MAC through FCS
- If the L3 packet is smaller than 46 bytes, the sender adds **padding bytes**
- Padding is usually **zeroes**
- **FCS is not part of the payload** – it is the trailer

> **Exam trap:** A series of `00000000` bytes at the end of the Ethernet payload is usually **padding**, not FCS and not part of the ping data.

3. ARP – Address Resolution Protocol

> **ARP maps an IPv4 address to a MAC address on the local network.**

Why ARP is needed

A host knows the destination IP address, but to build an Ethernet frame it needs the destination MAC address.

ARP Request

- Sent when a host needs to learn the MAC address for an IP
- **Destination MAC = broadcast:** `FF:FF:FF:FF:FF:FF`
- Sent to all hosts on the local network
- Contains: "Who has IP `X.X.X.X`? Tell me your MAC."

ARP Reply

- Sent by the host that owns the target IP
- **Destination MAC = unicast** to the requester
- Contains: “IP `X.X.X.X` is at MAC `YYYY.YYYY.YYYY`.”

ARP process summary

1. Host A wants to send to Host B’s IP.
2. Host A checks its ARP table.
3. If no entry exists, Host A sends an **ARP Request** broadcast.
4. Host B replies with an **ARP Reply** unicast.
5. Host A stores IP-to-MAC mapping in its **ARP table**.
6. Host A can now build the Ethernet frame.

> **Exam trap:** ARP Request = broadcast. ARP Reply = unicast.

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4. ARP Table

> **ARP table / ARP cache = IP address → MAC address mapping.**

- Used by hosts, routers, and Layer 3 devices
- Dynamic entries age out
- Static entries can be configured manually
- On Cisco devices: `show arp`
- On Windows PCs: `arp -a`
- Clear ARP cache on Cisco: `clear arp-cache`

PC `arp -a` output fields

Field	Meaning
Internet Address	IP address
Physical Address	MAC address
Type	Dynamic / Static

> **Exam trap:** “Internet Address, Physical Address, Type” is **ARP table output on a PC**, not Cisco switch MAC address table output.

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5. Ping / ICMP

> **Ping tests Layer 3 reachability using ICMP.**

ICMP messages used by ping

Message	Direction
ICMP Echo Request	Sent by the pinging device
ICMP Echo Reply	Sent by the target

Important behavior

- Ping uses IP addresses
- Before the first ICMP Echo Request can be sent, the host may need to perform ARP
- First ping may fail/time out while ARP resolves
- Later pings usually succeed
- Wireshark can show ARP followed by ICMP

Useful Wireshark filters

```
arp
icmp
arp || icmp
```

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6. MAC Address Table

> **MAC address table / CAM table = MAC address → switch port / VLAN mapping.**

A switch learns MAC addresses by examining the **source MAC** of incoming frames.

Cisco command

```
show mac address-table
```

Output fields

Field	Meaning
VLAN	VLAN associated with the MAC entry
MAC Address	Learned MAC address
Type	DYNAMIC / STATIC
Ports	Switch interface

> **Exam trap:** Cisco `show mac address-table` output includes **VLAN, MAC Address, Type, Ports**. It does **not** include IP addresses.

Dynamic MAC entries

- Learned automatically from source MAC addresses
- Age out after the default aging time — commonly **300 seconds / 5 minutes**
- Can be cleared manually

Clear MAC address table commands

```
clear mac address-table dynamic
clear mac address-table dynamic interface interface-id
clear mac address-table dynamic vlan vlan-id
clear mac address-table dynamic address mac-address
```

> **Quiz answer command:** `clear mac address-table dynamic interface interface-id`

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7. Switch Forwarding Behavior

Frame type	Switch behavior
Known unicast	Forward out the specific port
Unknown unicast	Flood out all ports except ingress
Broadcast	Flood out all ports except ingress
Multicast	May flood unless multicast snooping/control is used

> **Exam trap:** A switch floods **broadcast** and **unknown unicast** frames out all interfaces except the one it received the frame on.

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8. Hands-On Demo Notes

- GNS3 used to build a small switched network
- Ping used to generate traffic
- Wireshark used to inspect ARP and ICMP
- Clear MAC address table to observe relearning/flooding
- First ping may show ARP Request/Reply before ICMP Echo Request/Reply

Typical capture order

1. ARP Request — broadcast
2. ARP Reply — unicast
3. ICMP Echo Request
4. ICMP Echo Reply

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9. Quiz Questions + Answers

Q1. You send a 36-byte ping and see a long series of `00000000` at the end of the Ethernet payload. What are these zeroes?

Answer: b) They are padding bytes.

Why: Ethernet requires a minimum payload size of 46 bytes. If the Layer 3 packet is too small, padding zeroes are added. FCS is a trailer, not payload.

Q2. Which message is sent to all hosts on the local network?

Answer: a) ARP request

Why: ARP Requests use the broadcast MAC address `FF:FF:FF:FF:FF:FF`. ARP Replies and ICMP messages are normally unicast.

Q3. Which fields are present in `show mac address-table` output on a Cisco switch?

Answer: c) VLAN, MAC Address, Type, Ports

Why: Cisco switch MAC address table output includes VLAN, MAC, Type, and Ports. "Internet Address, Physical Address, Type" is ARP table output on a PC.

```
SW1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
   1    0c2f.b011.9d00    DYNAMIC     Gi0/0
   1    0c2f.b06a.3900    DYNAMIC     Gi0/2
Total Mac Addresses for this criterion: 2
SW1#
```

Q4. Which types of frames does a switch send out of all interfaces except the one it received the frame on?

Answer: a) Broadcast, unknown unicast

Why: Switches flood broadcast and unknown unicast frames. Known unicast frames are forwarded only out the correct port.

Q5. Which command clears all dynamic MAC addresses on a specific interface from the MAC address table?

Answer: d) `clear mac address-table dynamic interface interface-id`

Why: This clears dynamic MAC entries learned on the specified interface. Static entries are not removed by this command.

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10. Key Terms Glossary

Term	Meaning
Ethernet frame	Layer 2 PDU
FCS	Frame Check Sequence — error-detection trailer
Padding	Zero bytes added to meet minimum Ethernet payload size
ARP	Maps IPv4 address to MAC address
ARP Request	Broadcast message asking "Who has this IP?"
ARP Reply	Unicast message giving the MAC for an IP
ARP table	IP-to-MAC mapping table
ICMP Echo Request	Ping request
ICMP Echo Reply	Ping reply
MAC address table	MAC-to-port/VLAN mapping table on a switch
Unknown unicast	Frame with destination MAC not yet in MAC table
Flooding	Sending frame out all ports except ingress

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11. Exam Traps / Quick Review

- Ethernet minimum payload = **46 bytes**
- Minimum Ethernet frame = **64 bytes**
- Padding zeroes are **not FCS**
- ARP Request = **broadcast**
- ARP Reply = **unicast**
- ARP table maps **IP → MAC**
- MAC address table maps **MAC → port/VLAN**
- Switch floods **broadcast + unknown unicast**
- Known unicast is forwarded out one port
- Cisco MAC table fields: **VLAN, MAC Address, Type, Ports**
- PC ARP table fields: **Internet Address, Physical Address, Type**

- Clear dynamic MAC on interface: `clear mac address-table dynamic interface interface-id`

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12. Commands to Memorize

```
show mac address-table
show mac address-table dynamic
show mac address-table interface interface-id
clear mac address-table dynamic
clear mac address-table dynamic interface interface-id
show arp
clear arp-cache
ping <ip-address>
```

Wireshark filters:

```
arp
icmp
arp || icmp
```